Smart Chilli Crop Intelligence System Using an Autonomous Ground Rover

R26-IT-075

Our Team



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Introduction to the overall project

- The project develops a Smart Chilli Crop Intelligence System using an autonomous ground rover.
- Chilli cultivation requires accurate monitoring of soil, moisture, disease, stress and growth conditions.
- The rover collects field data through cameras, GNSS/IMU sensors and soil sensors.
- AI and machine learning models analyse crop health, soil status, irrigation needs and field zones.
- The final dashboard provides farmers and agronomists with clear, practical crop management recommendations.

Research Problem

- Chilli cultivation in Sri Lanka is highly affected by disease, water stress, soil imbalance and uneven field conditions.
- Farmers still depend mainly on manual observation, fixed irrigation schedules and delayed soil testing.
- Early disease symptoms, nutrient deficiencies and moisture stress are often missed until crop damage has already occurred.
- Ground rover navigation can also become unreliable when GPS/GNSS signals degrade under crop canopy, trees or field obstructions.
- Existing solutions usually handle disease detection, irrigation, soil analysis or mapping separately, rather than as one integrated system.

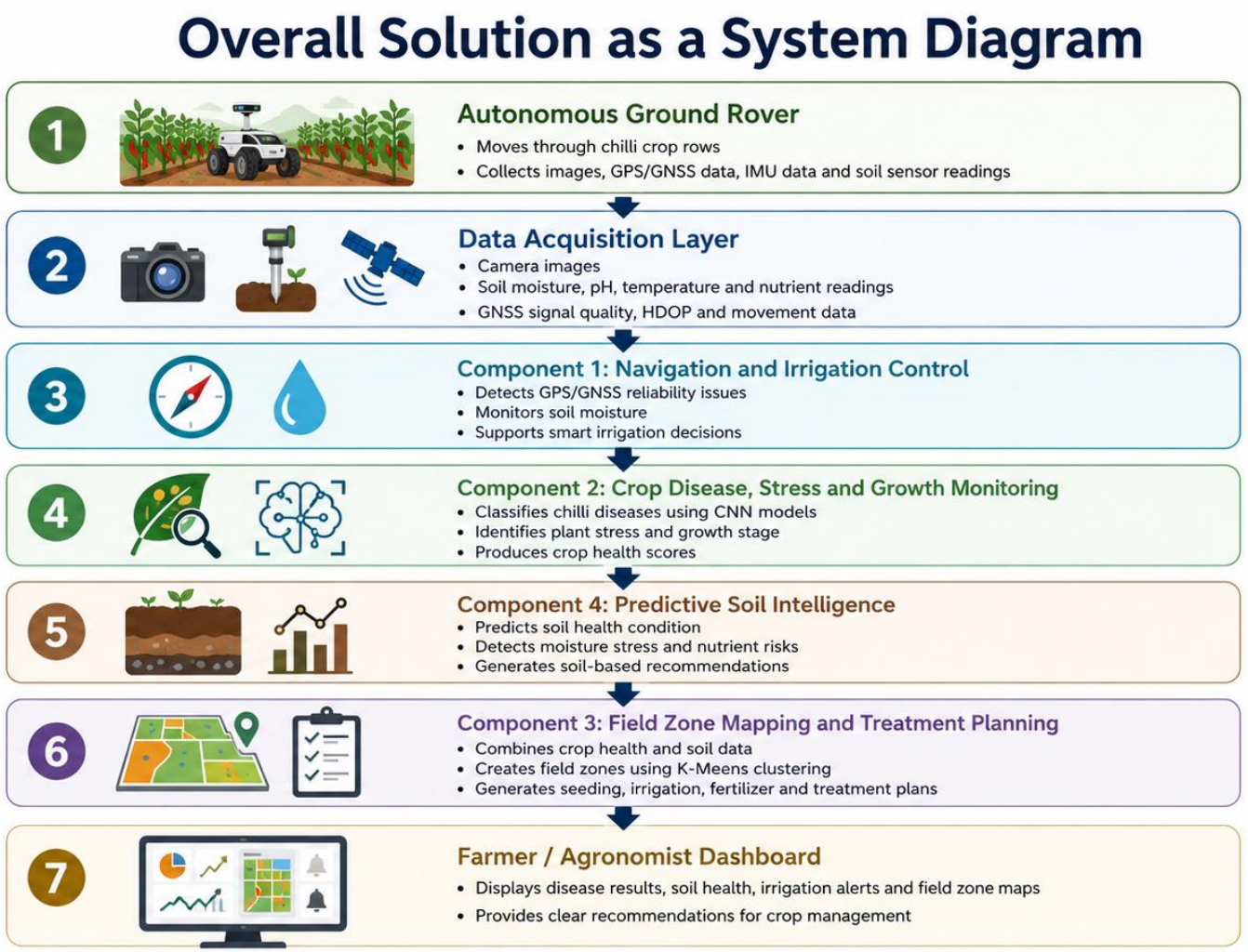
Research Question

How can an autonomous ground rover-based intelligent system improve chilli crop monitoring and decision-making by integrating disease detection, soil health prediction, irrigation control, and field zone mapping?

Research Objectives

- To design and develop a Smart Chilli Crop Intelligence System using an autonomous ground rover.
- To monitor rover navigation reliability and support sensor-based drip irrigation decisions.
- To detect chilli crop diseases, stress conditions and growth stages using AI-based image analysis.
- To predict soil health conditions using real-time soil sensor readings and machine learning models.
- To divide the chilli field into meaningful management zones using spatial clustering techniques.
- To generate zone-specific recommendations for irrigation, fertilizer use, seeding density and disease treatment.
- To present all results through a farmer-friendly dashboard for better decision-making.

Overall System Diagram





Intelligent Navigation and Sensor-Based Irrigation Control

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Research Question, Specific and Sub Objectives

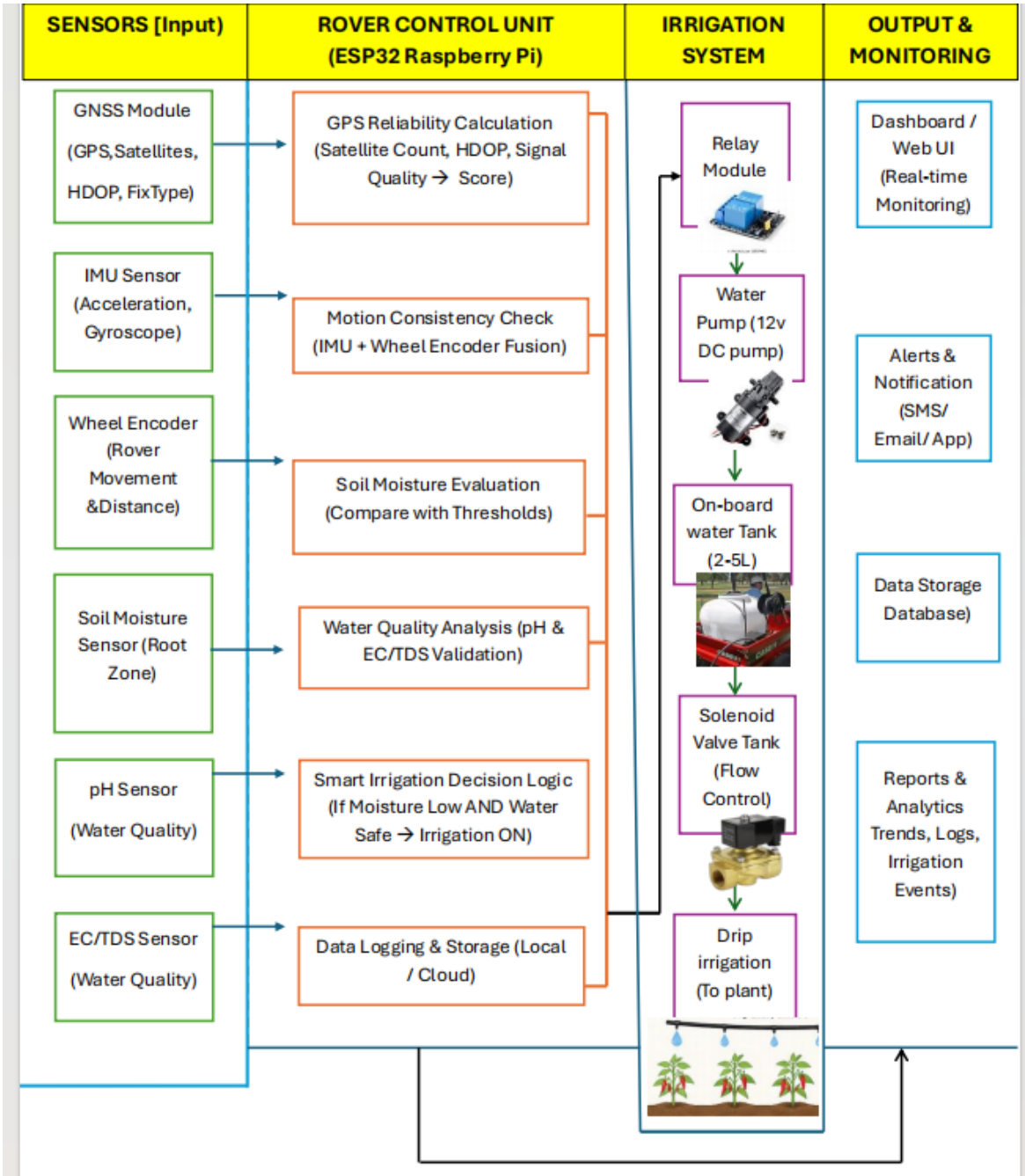
Sub-Research Question

How can a low-cost autonomous ground rover improve chilli field navigation reliability and support smart sensor-based irrigation control?

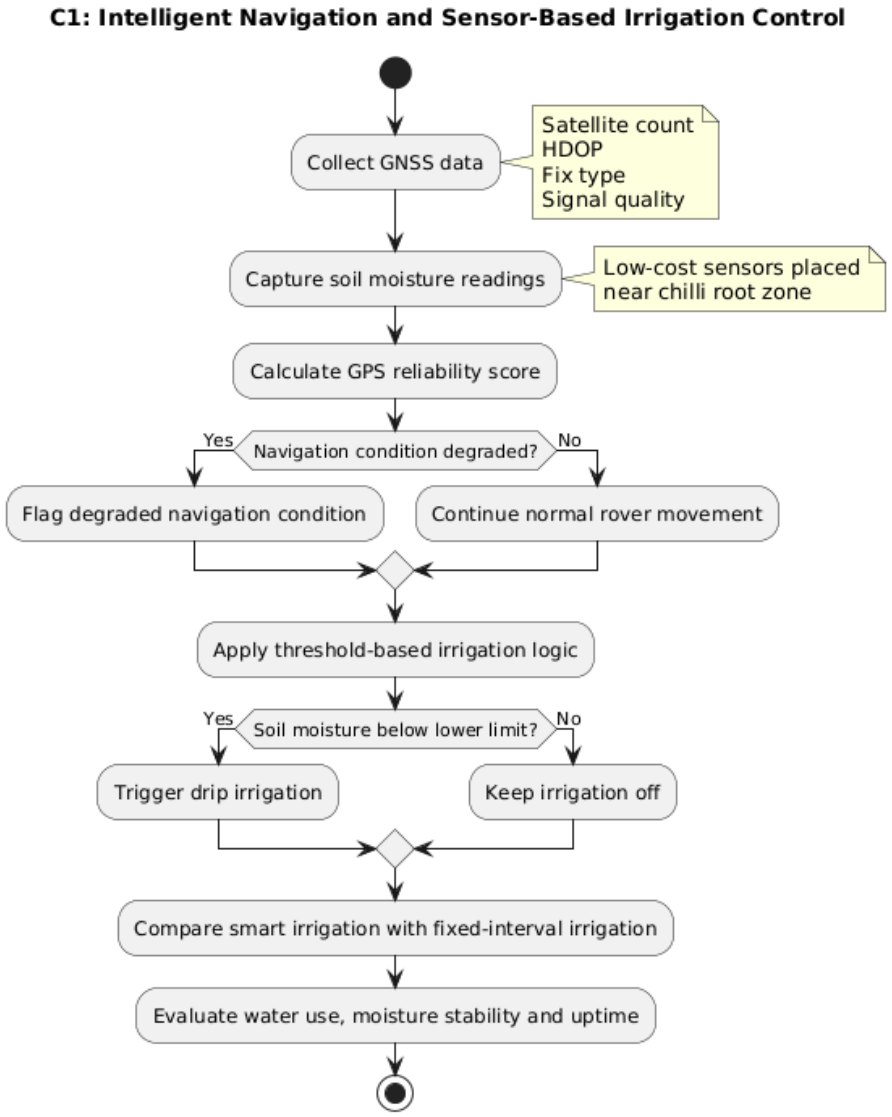
Sub-Objectives

- ❑ Define a real-time GPS/GNSS reliability score using satellite count, HDOP and signal quality.
- ❑ Develop a soil-moisture sensing module for chilli field monitoring.
- ❑ Implement threshold-based drip irrigation control using lower and upper moisture limits.
- ❑ Compare smart irrigation with fixed-interval irrigation using water use and moisture stability.
- ❑ Provide logs, dashboard outputs and data feeds for Components 2, 3 and 4.

Component Diagram

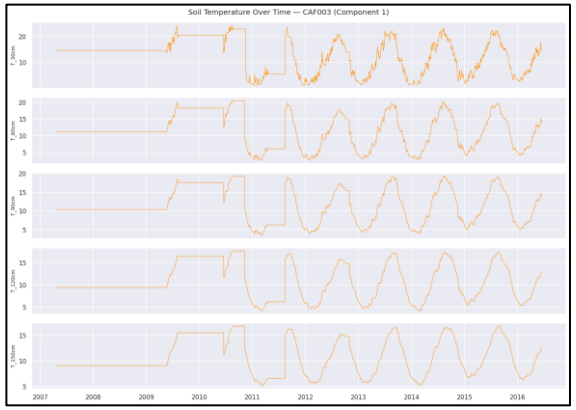


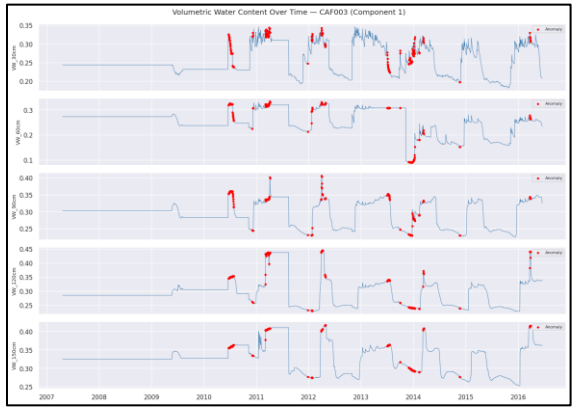
Methodology

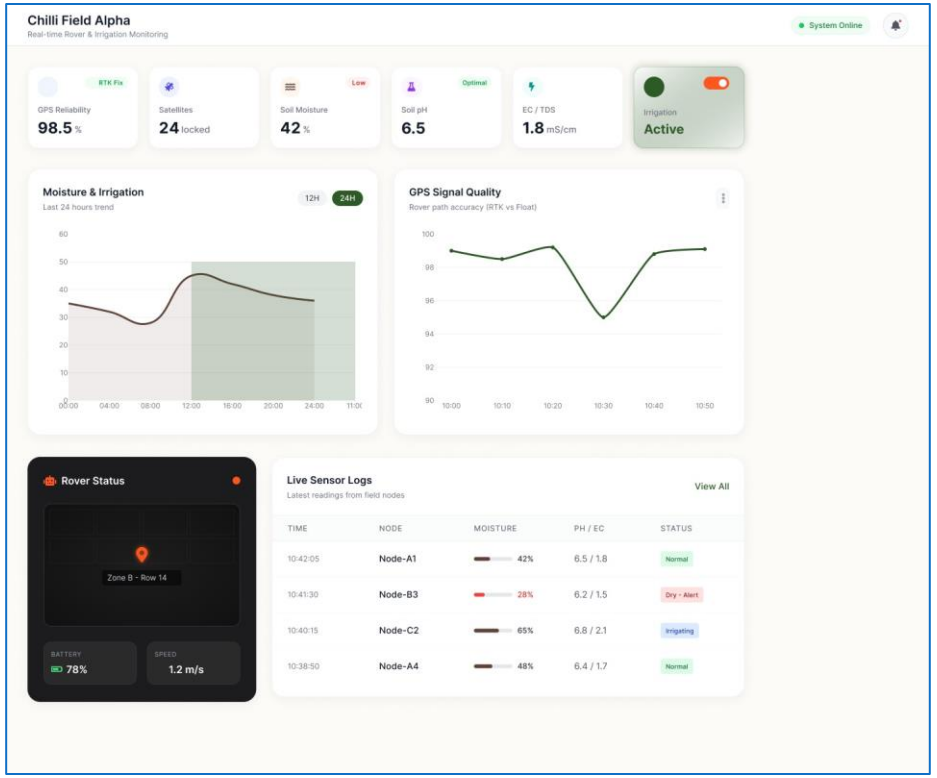


Completion of the project

- Soil moisture anomaly detection notebook was developed and tested.
- Sensor-based data processing was implemented using Python and machine learning.
- GNSS reliability and irrigation logic were designed for rover field operation.
- Dashboard (with UI) integration was prepared to display sensor logs and irrigation-related outputs.
- Evaluation metrics were defined for detection rate, false alarms, alert latency, water saving and uptime.



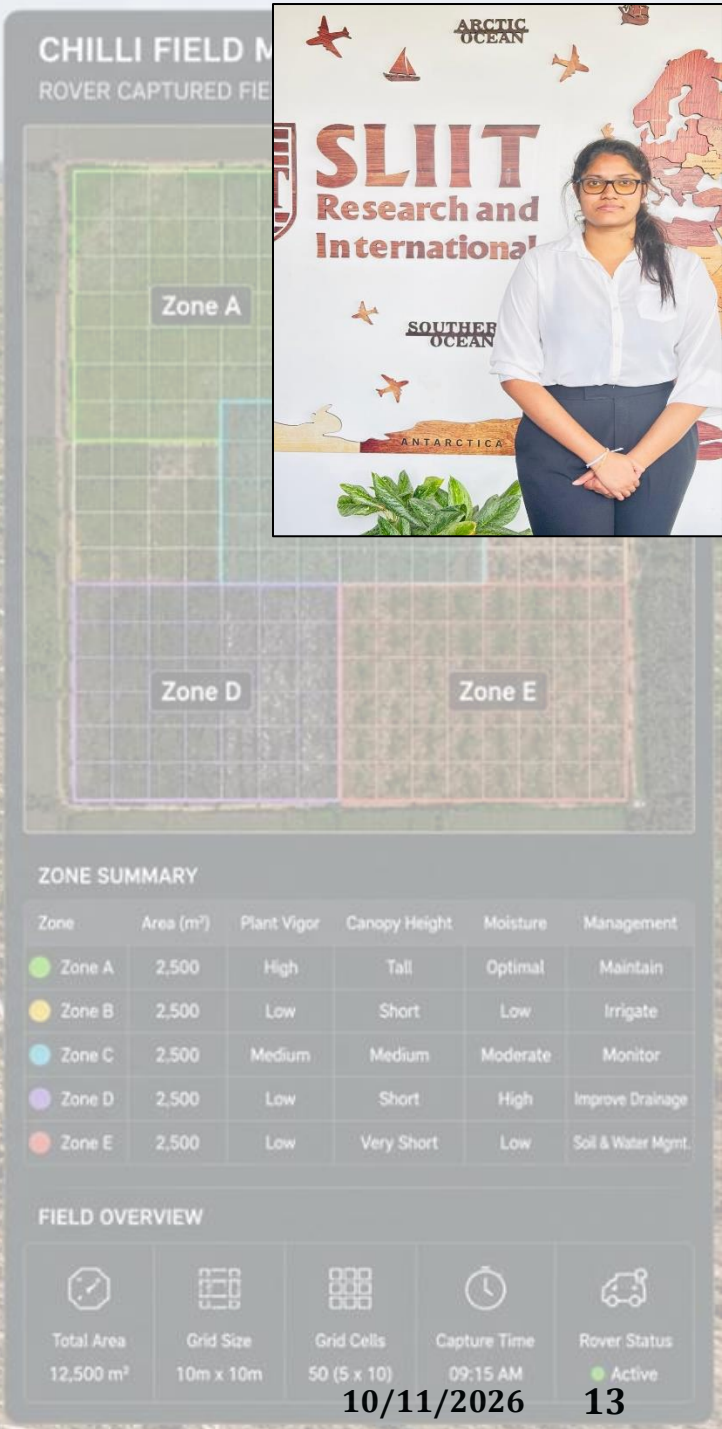




AI-Based Chilli Crop Disease, Stress and Growth Monitoring

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Research Question, Specific and Sub Objectives

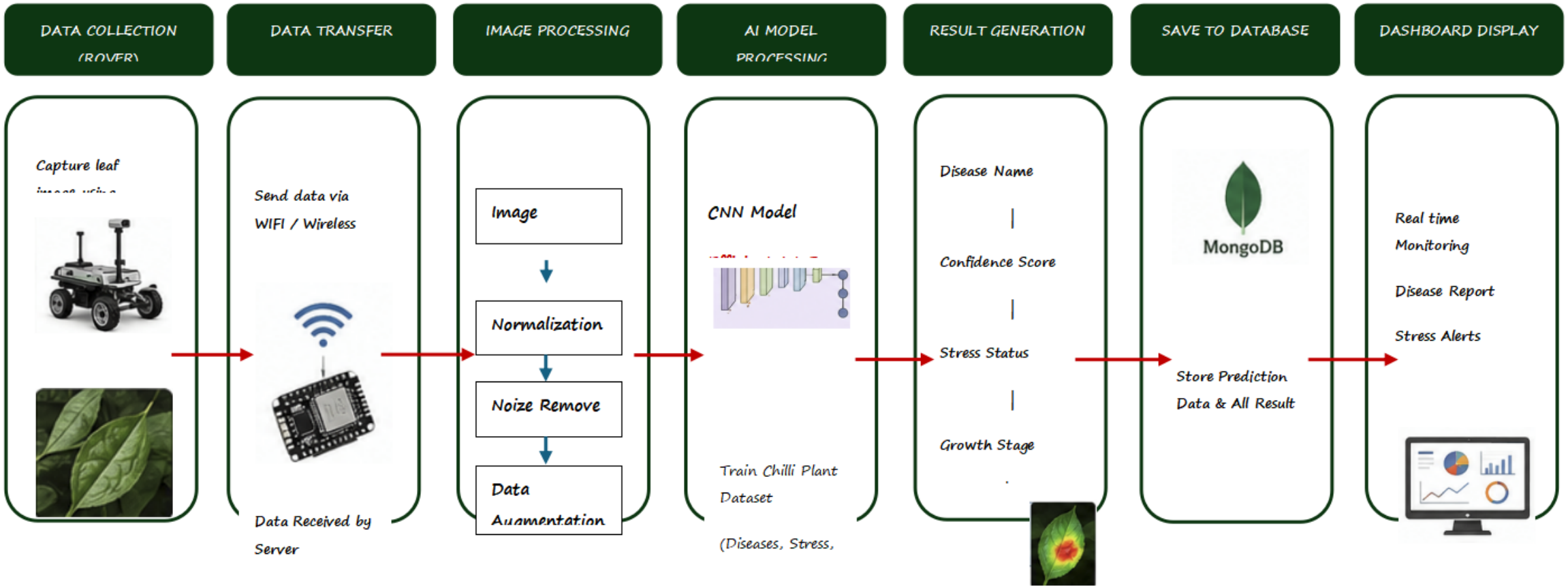
Sub-Research Question

How can an AI-based ground rover system detect chilli crop diseases, stress conditions and growth stages using image analysis and sensor data?

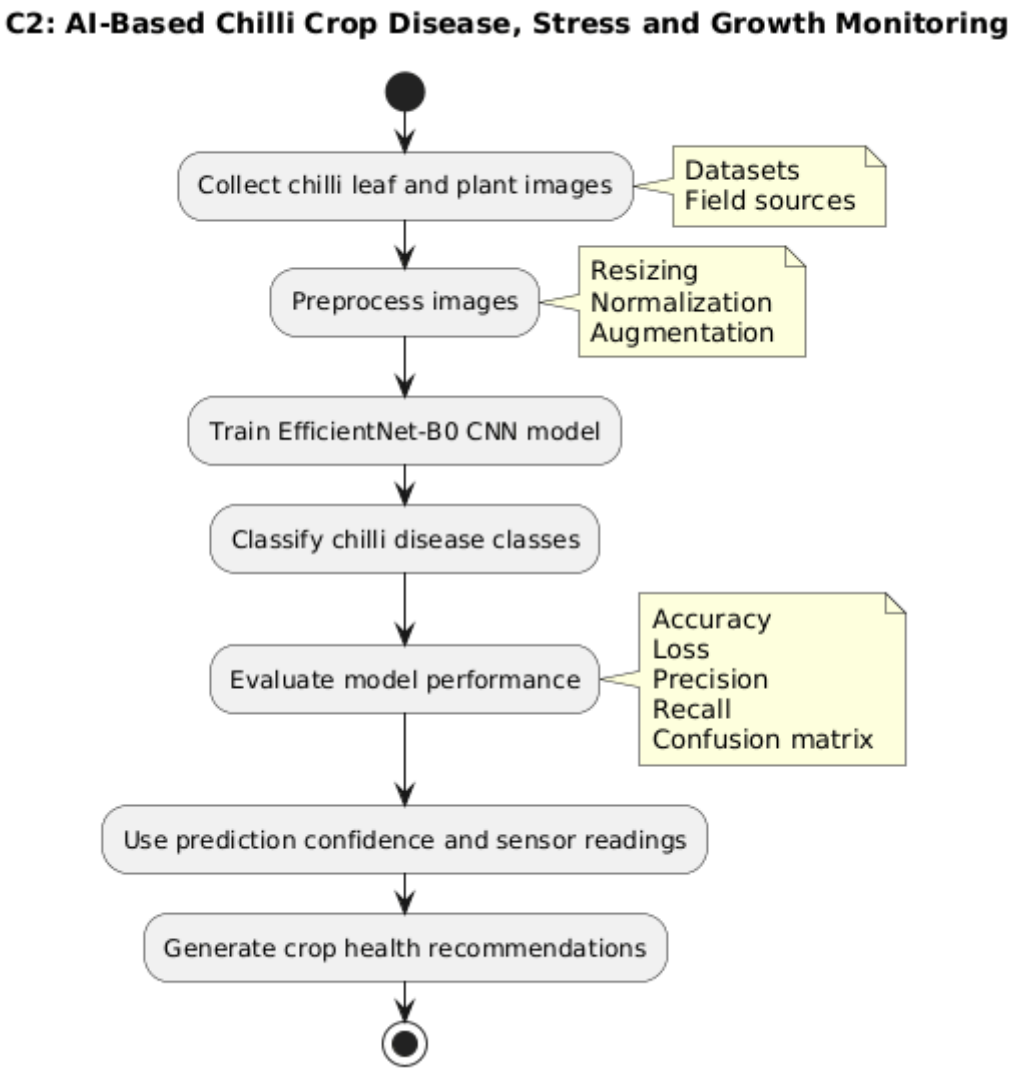
Sub-Objectives

- ❑ Develop an image classification model to identify major chilli crop diseases.
- ❑ Detect plant stress conditions using visual symptoms and environmental data.
- ❑ Classify chilli plant growth stages from rover-captured field images.
- ❑ Apply Grad-CAM explainability to show disease-affected image regions.
- ❑ Generate crop health outputs for field zone mapping and soil intelligence components.

Component Diagram

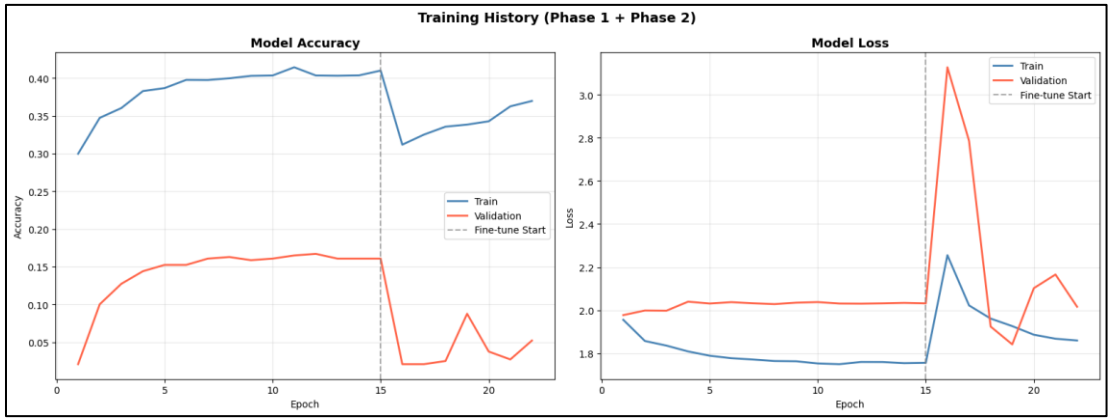


Methodology



Completion of the project

- Chilli disease detection notebook was developed and tested using image datasets.
- EfficientNet-B0 based CNN model was implemented for disease classification.
- Image preprocessing, augmentation and dataset splitting were completed.
- Model evaluation outputs were generated using accuracy, loss and confusion matrices.
- Disease prediction results were prepared for dashboard and component integration.







Field Zone Mapping and Autonomous Seeding and Treatment Planning

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Research Question, Specific and Sub Objectives

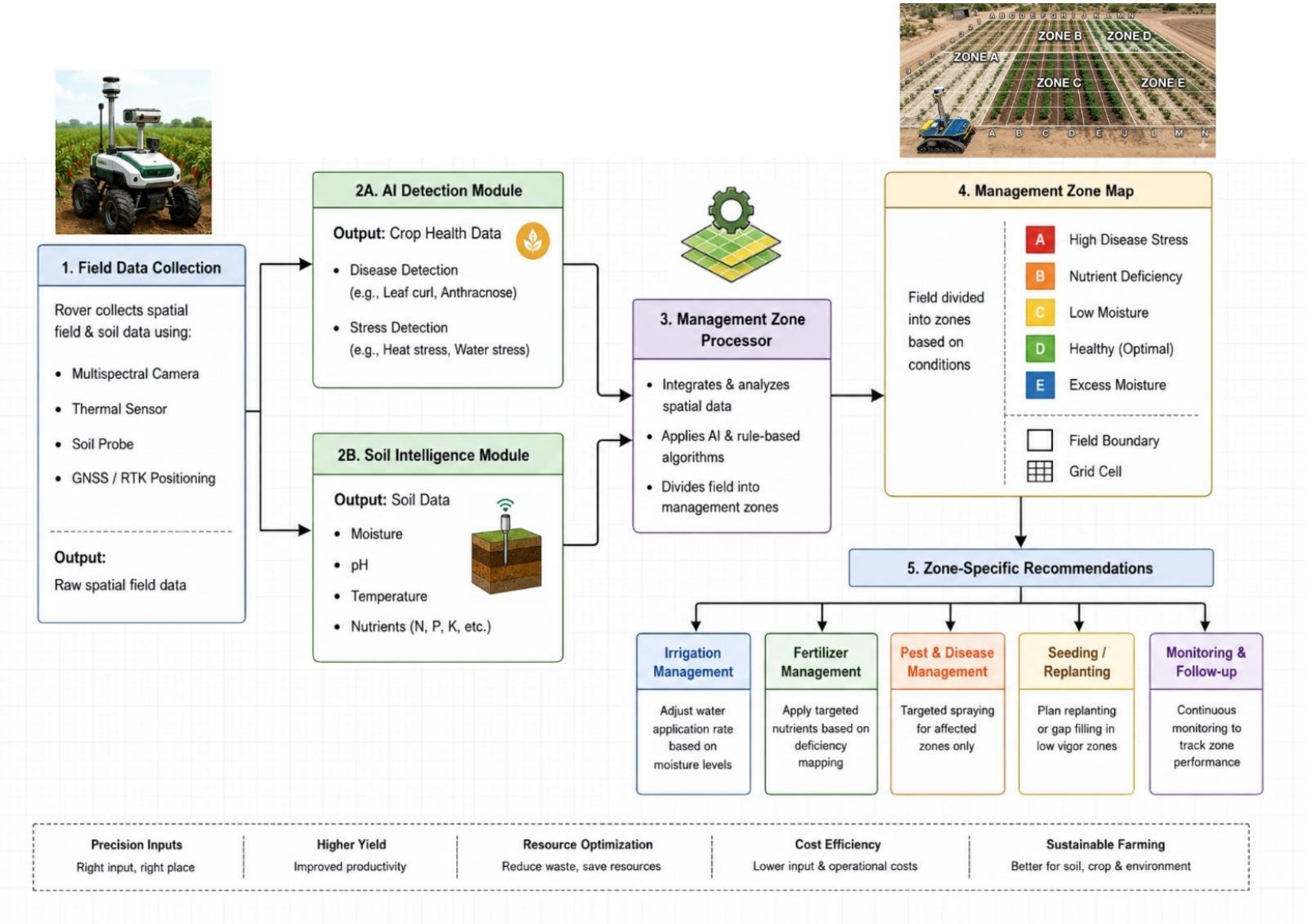
Sub-Research Question

How can rover-collected crop health and soil data be used to map chilli field zones and generate autonomous seeding and treatment plans?

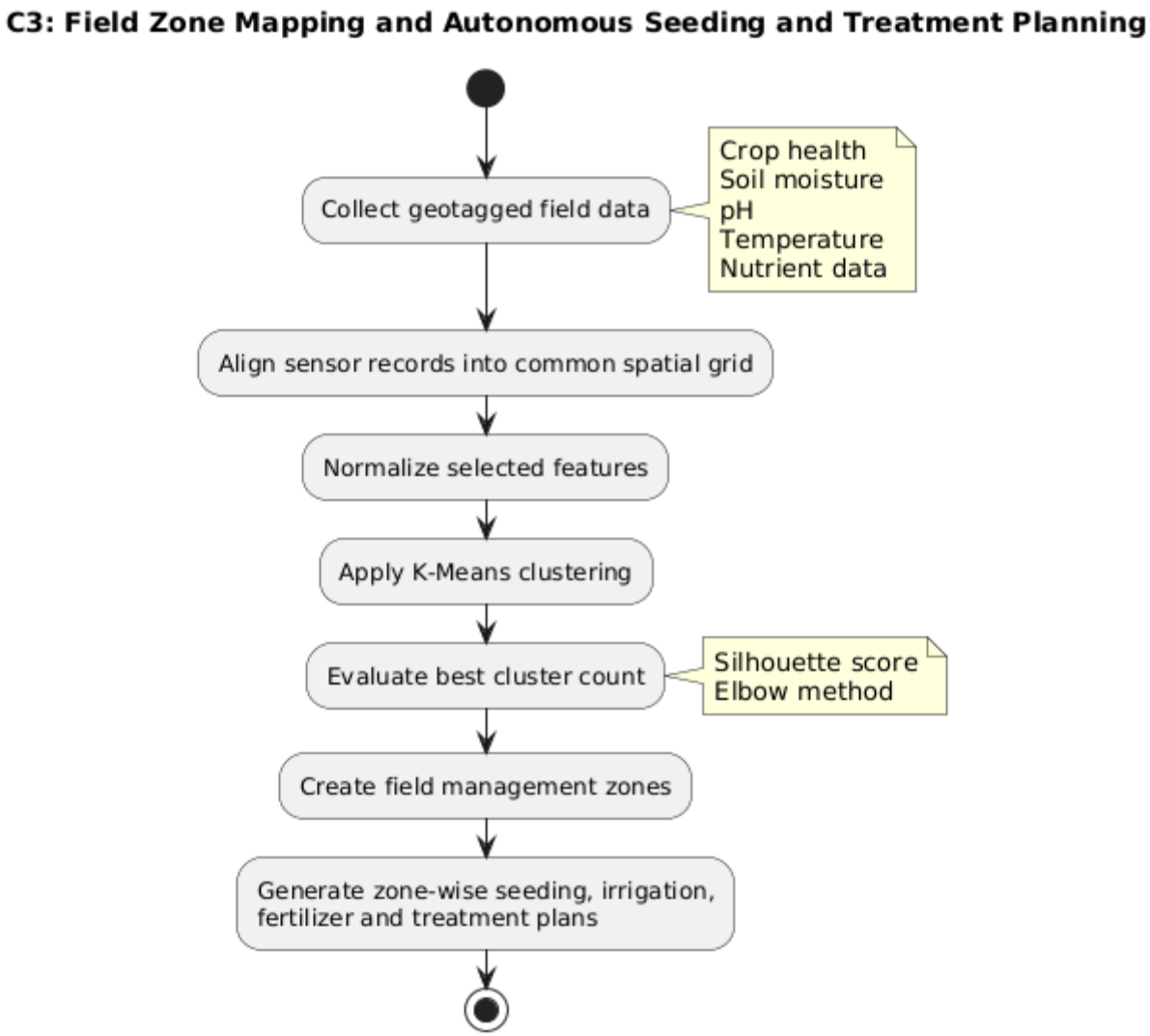
Sub-Objectives

- ❑ Integrate crop health outputs and soil sensor readings into one spatial dataset.
- ❑ Apply K-Means clustering to divide the chilli field into management zones.
- ❑ Classify zones as healthy, moderately stressed or critically stressed.
- ❑ Generate seeding, irrigation, fertilizer and treatment recommendations for each zone.
- ❑ Display field zones and recommendations through a GIS-based dashboard.

Component Diagram

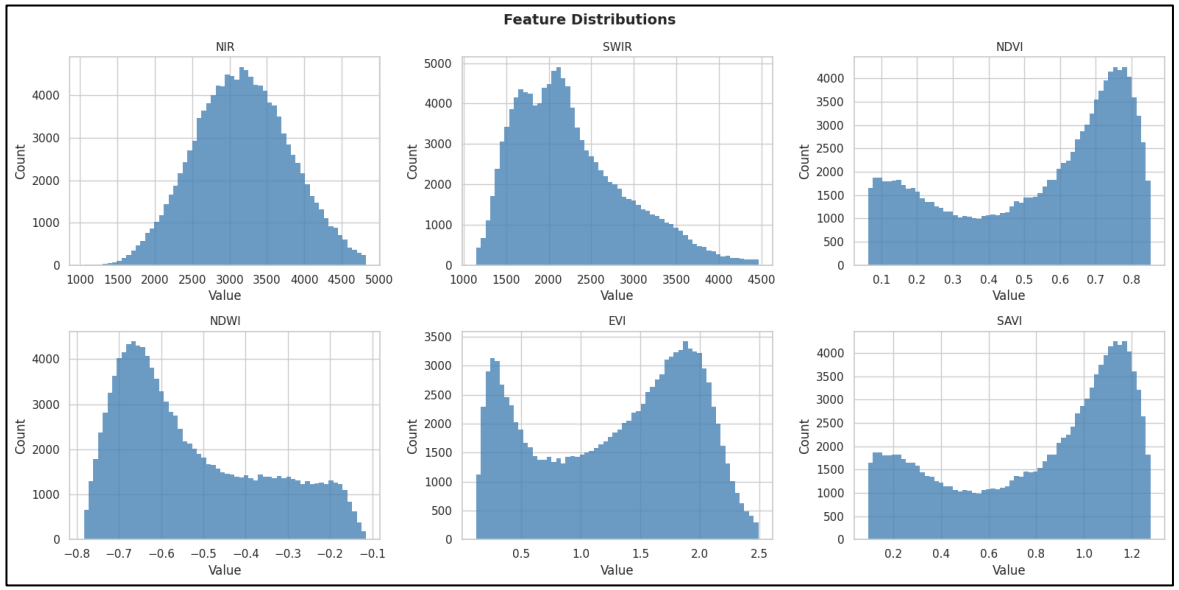


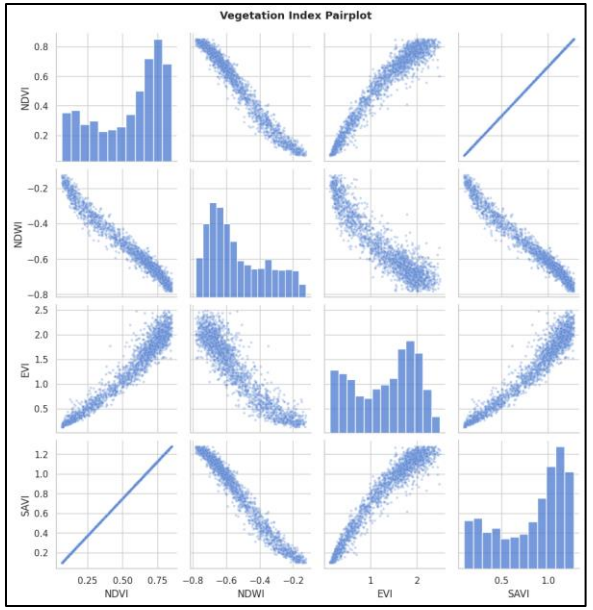
Methodology



Completion of the project

- Field zone mapping notebook was developed and tested using spatial data.
- K-Means clustering was implemented for automatic zone identification.
- Cluster evaluation was completed using silhouette and clustering performance metrics.
- Zone classification logic was prepared for healthy, moderate and critical field areas.
- Dashboard-ready outputs were prepared for treatment planning and map visualization.







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Research Question, Specific and Sub Objectives

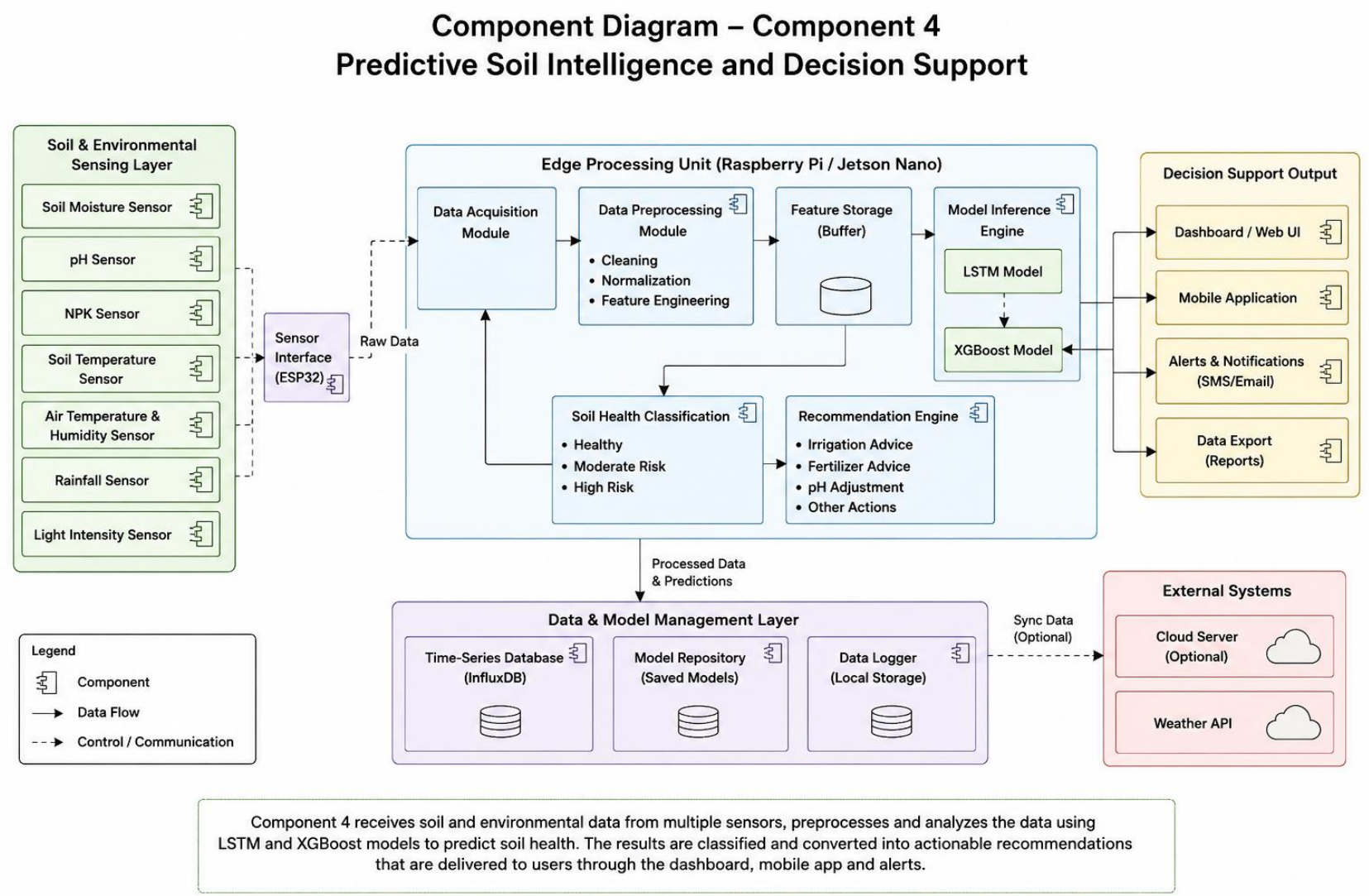
Sub-Research Question

How can rover-collected soil sensor data be used to predict soil health conditions and support better chilli cultivation decisions?

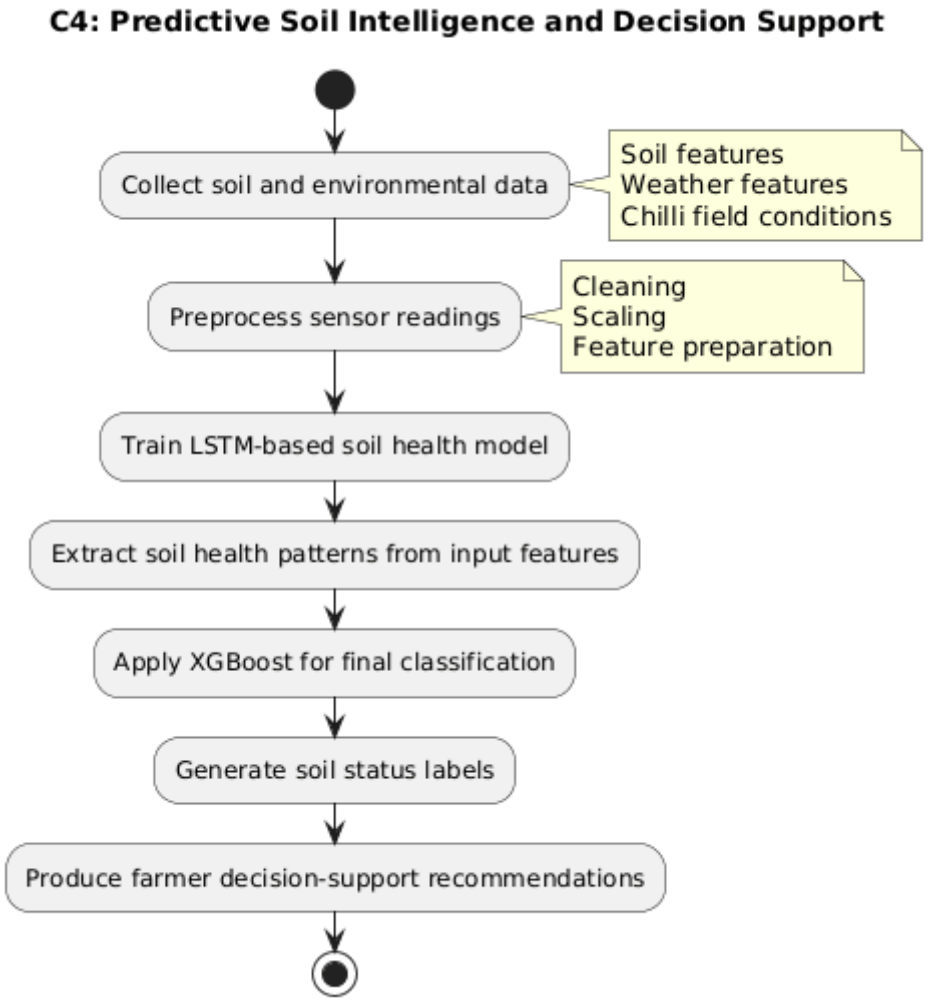
Sub-Objectives

- ❑ Collect pH, temperature and nutrient-related readings from the rover.
- ❑ Develop predictive models to classify and forecast soil health conditions.
- ❑ Identify cultivation risks such as moisture stress, nutrient depletion and pH imbalance.
- ❑ Combine soil intelligence with crop health outputs from Component 2.
- ❑ Generate farmer-friendly recommendations for irrigation, fertilizer and field intervention.

Component Diagram

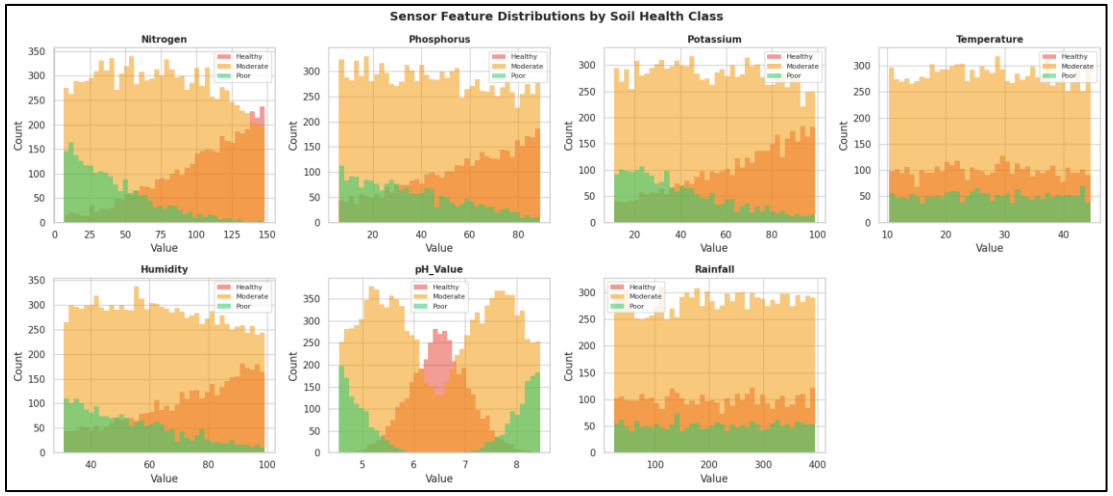


Methodology



Completion of the project

- Soil health detection notebook was developed and tested using agricultural sensor data.
- LSTM and XGBoost models were implemented for soil health classification.
- Soil features such as N, P, K, temperature, humidity, pH and rainfall were processed.
- Model performance was evaluated using accuracy and classification outputs.
- Soil health results were prepared for dashboard display and decision support integration.





Solution Approach & User Requirements

How the Solution Addresses the Sub-Problems

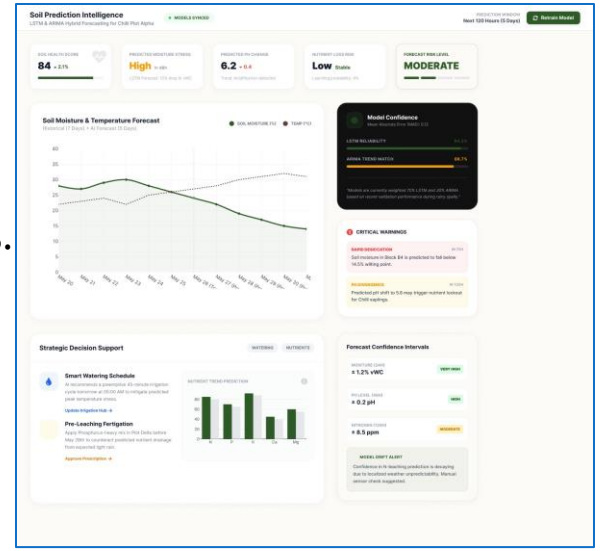
- Predicts soil health conditions using ML-based analysis.
- Forecasts nutrient status and soil quality trends.
- Combines sensor data and environmental variables for decision support.
- Detects healthy, moderate, and poor soil conditions.
- Supports long-term sustainable chilli cultivation planning.

- **User Requirements Addressed**
- Farmers need soil quality monitoring.
- Need for nutrient deficiency identification.
- Requirement for data-driven fertilizer recommendations.
- Need for improved crop productivity.
- Requirement for predictive agricultural analytics.

- **Technologies / Design Contribution**
- LSTM for soil trend forecasting.
- XGBoost for soil classification.
- Multi-sensor feature fusion.
- Real-time decision-support analytics dashboard.

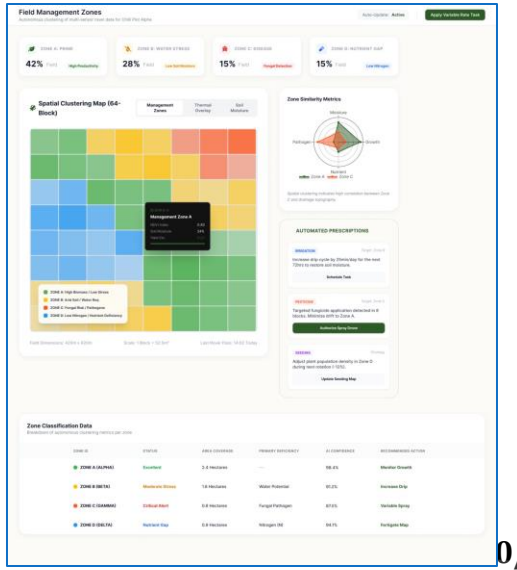
Prototype Design & User Feedback

- **Prototype / Design Excellence**
- Built soil intelligence prediction pipeline.
- Integrated Nitrogen, Phosphorus, Potassium, pH, humidity, and rainfall analysis.
- Achieved high classification accuracy using XGBoost and LSTM models.



- **User Feedback on Prototype**
- **Farmers**
- Helpful for understanding soil condition changes.
- Easier fertilizer planning process.

- **Agricultural Experts**
- Positive response to soil classification accuracy.
- Recommended adding micronutrient analysis.



- **Expected Outcomes**
- Improved soil health management.
- Better nutrient optimization.
- Increased chilli cultivation productivity.